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STRUCTURAL DEPENDENCE OF THE STATE ON CAPITAL

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A central claim of both Marxist and neoclassical political theory is that under capitalism all governments must respect and protect the essential claims of those who own the productive wealth of society. This is the theory of "structural dependence of the state on capital." Using a formal model, the internal logic and the robustness of the theory is examined. We conclude that in a static sense the theory is false: virtually any distribution of consumption between wage earners and owners of capital is compatible with continual private investment once an appropriate set of taxes and transfers is in place. Yet the state may be structurally dependent in a dynamic sense. Policies that, once in place, redistribute income without reducing investment do reduce investment during the period in which they are anticipated but not yet implemented.

Capitalism is a system in which many scarce resources are owned privately, and decisions about allocating them are a private prerogative. Democracy is a system through which people as citizens may express preferences about allocating resources that they do not privately own. Hence the perennial question of political theory and of practical politics concerns the competence of these two systems with regard to each other. Is it possible for governments to control a capitalist economy? In particular, is it possible to steer the economy against the interests and preferences of those who control productive wealth?

The central and only distinctive claim of Marxist political theory is that under capitalism all governments must respect and protect the essential claims of those who own the productive wealth of society. Capitalists are endowed with public power, power which no formal institutions can overcome (Luxemburg 1970; Pashukanis 1951). People may have political rights, they may vote, and governments may pursue popular mandates. But the effective capacity of any government to attain whatever are its goals is circumscribed by the public power of capital. The nature of political forces that come into office does not alter these limits, it is claimed, for they are structural—a characteristic of the system, not of the occupants of governmental positions nor of the winners of elections.

During the past 20 years Marxists have developed several theories to explain why all governments in capitalist societies are bound to act in the interests of capitalists. In one explanation, state managers internalize the goals of capitalists and use the state as an instrument on their behalf (Miliband 1969). Another explanation emphasizes the generic structural and functional limitations: under capitalism, the state cannot organize production (Offe 1975), mandate investment (Lind-

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blom 1977), or command consumption, because these are prerogatives reserved to owners of private property. But the most daring, because the least contingent, theory claims that it does not matter who the state managers are, what they want, and whom they represent. Nor does it matter how the state is organized and what it is legally able or unable to do. Capitalists do not even have to organize and act collectively: it suffices that they blindly pursue narrow, private self-interest to sharply restrict the options of all governments. This is the theory of "structural dependence of the state on capital."

This theory begins with the hypothesis that the entire society depends on the allocation of resources chosen by owners of capital. Investment decisions have public and long-lasting consequences: they determine the future possibilities of production, employment, and consumption for all. Yet they are private decisions. Since every individual and group must consider its future, since future consumption possibilities depend on present investment, and since investment decisions are private, all social groups are constrained in the pursuit of their material interests by the effect of their actions on the willingness of owners of capital to invest, which in turn depends on the profitability of investment. In a capitalist society, the trade-off between present and future consumption for all passes through a trade-off between consumption of those who don't own capital and profits (Przeworski 1985, chap. 5).

Consider this dependence from the point of view of one group, wage earners. At any instant of time, wages and profits are indeed inversely related, as Marx argued in *Wage, Labour, and Capital* (1952a). In a world without a future, wage earners would be best-off consuming the entire product, indeed confiscating the capital stock. But wage earners care about their future as well as present income, and future wages depend on private investment. If firms respond to wage increases with less investment, wage earners may be best-off moderating their wage demands. Workers' future income depends upon the realization of capitalists' present interests.

While the theory is usually stated with regard to workers' wage demands, to the extent that material means are required to advance their welfare, structural dependence binds all groups: minorities struggling for economic equality, women wanting to transform the division of labor within the household, old people searching for material security, workers striving for safer working conditions, or the military seeking bombs. It is in this sense that capitalism is a class society: not in the sense that there are always two organized classes, but in the sense that the structure of property characteristic of capitalism makes everyone's material conditions dependent upon the private decisions of owners of wealth.

The theory of structural dependence continues with the inference that because the entire society depends on the owners of capital, so must the state. Whether particular governments have interests and goals of their own or they act on behalf of a coalition of groups or a class, the pursuit of any objectives that require material resources places governments in the situation of structural dependence. Politicians seeking reelection must anticipate the impact of their policies on the decisions of firms because these decisions affect employment, inflation, and personal income of voters: vote-seeking politicians are dependent on owners of capital because voters are. Even a government that was a perfect agent of wage earners could not and would not behave much differently from one that represents capitalists. If workers are best-off with a fair dose of wage restraint, a prolabor government will also avoid policies that dramatically alter the distribution of income and

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wealth. The range of actions that governments find best for the interests they represent is narrowly circumscribed, whatever these interests may be.

The reason the state is structurally dependent is that no government can simultaneously reduce profits and increase investment. Firms invest as a function of expected returns; policies that transfer income away from owners of capital reduce the rate of return and thus of investment. Governments face a trade-off between distribution and growth, between equality and efficiency. They can trade a more (or less) egalitarian distribution of income for less (or more) investment but they cannot alter the terms of this trade-off: this is the central thesis of the theory of structural dependence. Governments can and do choose between growth and income distribution; but because material welfare of any constituency depends upon economic growth as well as its share of income and because distribution can be achieved only at the cost of growth, all governments end up pursuing policies with limited redistributive effects.

At this moment the reader may remark that this is the neoliberal theory as well. It is. The Chicago school argues that all transfers of income cause deadweight losses. All support-maximizing governments, it is claimed, are tempered in their zeal for redistribution by the fact that owners of endowments will increasingly withdraw them from productive uses as taxes rise. (Bates and Lien 1985, Becker 1983, and Peltzman 1976 are presentations of formal models.) The difference between the two theories is that neoliberals are "pluralists"; that is, they are agnostic about the groups that have the power of inflicting the losses on the public by withdrawing their endowments. This difference should not obscure, however, the fact that both theories understand in the same way the relation between income distribution and investment.

The belief that under capitalism governments are structurally dependent on capital is widespread. Miliband (1969, 152) portrayed this dependence as follows: "Given the degree of economic power which rests in the 'business community' and the decisive importance of its actions (or its nonactions) for major aspects of economic policy, any government with serious pretensions to radical reform must either seek to appropriate that power or find its room for radical action rigidly circumscribed by the requirements of 'business confidence.' " Block (1977, 15) maintained that "in a capitalist economy the level of economic activity is largely determined by private investment decisions of capitalists. This means that capitalists, in their collective role as investors, have a veto over state policies in that their failure to invest at adequate levels can create major political problems for state managers." Lindblom (1977, 172, 175) observed that "because public functions in the market system rest in the hands of businessmen, it follows that jobs, prices, production, growth, the standard of living, and the economic security of everyone all rest in their hands. . . . In the eyes of government officials, therefore, businessmen do not appear simply as representatives of a special interest. . . . They appear as functionaries performing functions that government officials regard as indispensable." Offe (1975, 234) analyzed the predicament of the state vis-à-vis the private economy: "The political system can only make offers to external, autonomous bodies responsible for decisions: either these offers are not accepted, thus making the attempts at direction in vain, or the offers are so attractive in order to be accepted that the political direction for its part loses its autonomy because it has to internalize the aims of the system to be directed."

The theory of structural dependence can be evaluated in three ways. One

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method is to examine differences in policy choices and outcomes under different governments. Unfortunately, while there is much evidence that some policy outcomes covary with partisan orientations of governments, these findings cast little light on the existence of structural constraints that bind all governments. We cannot know whether the observed differences exhaust the realm of possibility. (The classical study was Hibbs 1977. See Cameron 1984, Castles 1982, Maravall 1979, and Shalev 1983 for literature reviews.) The second method is to examine the limiting cases: the experience of governments that came into office with programs of major transformations, in particular, programs of simultaneous nationalization of some industries and redistribution of income (Kolm 1977). All such governments either compromised their original program, including total reversals, like that committed by the French Socialists, or provoked profound economic crises and were overthrown by force, as was the government of the Unidad Popular in Chile. These experiences prompted Rueschemeyer and Evans (1985, 62) to observe that "even in countries where the state appears to be in the strongest position relative to private capital, . . . the state remains dependent on private capital, foreign and domestic, not only to promote accumulation but also to produce a surplus in which the state itself may share. The strict limits under which the state must operate in a dependent capitalist economy are grimly indicated by the severe problems confronted by social democratic regimes such as those of Salvador Allende in Chile and Michael Manley in Jamaica." Yet again, these experiences cannot speak to the issue of limits and possibilities: one can always cite some counterfactual actions that would have perhaps avoided the disasters. The issue concerns possibility and possibilities cannot be determined on the basis of limited historical experience. Hence, finally, the appropriate method to evaluate the hypothesis of structural dependence is to construct a formal model with which the internal logic of the theory can be explored and the robustness of its conclusions examined.

The Theory Reconstructed

Assumptions

To examine the validity of this theory, we need some assumptions. Suppose the society is composed only of wage earners and owners of capital. Wage earners consume all of their incomes and own no capital. There are firms, which receive the income from capital, invest some part of it, and distribute the rest to the shareholders. Let the net national income at time τ be $Y(\tau)$, let aggregate wages be $W(\tau)$, let investment be $I(\tau)$, and let profits net of investment be $P(\tau)$. Then $W(\tau) + I(\tau) + P(\tau) = Y(\tau)$. Let us implicitly define the share of wages in the national product, m, as $W(\tau) = mY(\tau)$. The volume of investment is $I(\tau) = s(1 - m)Y(\tau)$, where s is the rate of investment out of the total return to capital, $(1 - m)Y$. The volume of profits net of investment is then $P(\tau) = (1 - s)(1 - m)Y(\tau)$.

We make the simplest possible assumption about production: output is a linear function of capital. Expressed in terms of growth, $Y'(\tau) = \nu I(\tau) = \nu s(1 - m)Y(\tau)$, $\nu > 0$, where ν is the constant productivity of capital. The parameter ν represents the state of technology—the amount of additional output that can be produced when an additional unit is added to the capital stock. Thus, output grows at the rate of $\nu s(1 - m)$ per year. For example, if $\nu = .25$, $s = .50$ and $m = .80$, the economy is growing at an annual rate of 2.5%.[1]

Firms make investment decisions. They are perfect agents of their homogeneous shareholders. Thus, firms maximize the utility that their shareholders derive from

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profits.[2] Shareholders derive their utility from present and discounted future consumption, where ϱ is the rate at which the future is discounted. For the sake of algebraic simplicity, we assume shareholders' utility in each period is given by a logarithmic utility function.[3] Let P^* represent the present value of shareholders' welfare: the value they attach today to a particular sequence of present and future consumption. Then P^* can be written

$$P^* = \int_0^\infty e^{-\varrho\tau} \ln[P(\tau)]d\tau, \varrho > 0. \qquad (1)$$

Finally, wage earners derive their utility from consumption in the same manner.[4] Wage earners' welfare is given by

$$W^* = \int_0^\infty e^{-\varrho\tau} \ln[W(\tau)]d\tau.$$

These assumptions are perhaps the simplest possible ones. Many can be complicated to introduce more descriptive realism without altering the qualitative conclusions. In the hope of remaining accessible to a wider audience, we assert the analytic results without proof in the text. Complete derivations are contained in the Appendix.

Structural Dependence without the State

Consider the economy without state intervention. The state is there: it enforces contracts, standardizes weights and measures, and the like, but it does not touch income or investment.

Firms choose a rate of investment to maximize P^* in Equation 1 given that the share of wages in income is m. The solution to this problem (for $m < 1$), as shown in the Appendix, is

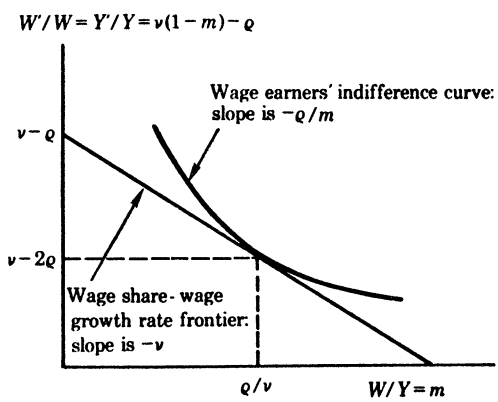
$$s^*(m) = 1 - \frac{\varrho/\nu}{1 - m} = s^M, \qquad (2)$$

where s^M stands for "market rate."[5] This is the investment rate firms will choose in the absence of taxes. Note that the rate of investment falls as the share of wages increases, $ds^*/dm < 0$.[6] Note also that the rate of investment is always less than 1, and it may be less than 0. Firms may find it optimal to disinvest. Indeed, as m approaches 1, s^M approaches negative infinity. Note, finally, that without taxes, shareholders are consuming the income share $(1 - s^*)(1 - m) = \varrho/\nu$, and firms are investing the share $I/Y = 1 - \varrho/\nu - m$ of national income. If $\varrho/\nu > 1$ or if the rate at which shareholders discount the future exceeds the productivity of capital, firms will disinvest, whatever the wage share.

Because shareholders always maintain their current consumption at ϱ/ν of income, any increase in the wage share must be exactly offset by a decline in the share of income going to investment: $d(I/Y)/d(W/Y) = -1$. Wage earners thus face a trade-off between the wage share and the rate of growth of national income (which determines the rate of growth of wages) as illustrated in Figure 1.

The straight line represents all possible combinations of the wage share and the

Figure 1. Wage Earners' Trade-off between Wage Share and Wage Growth



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rate-of-wage growth as income is transferred from wages to investment and vice versa. If wage earners were organized in a centralized union federation with sufficient power to set wages unilaterally, they would maximize W^* by choosing the point on the wage share–wage growth rate frontier that is tangent to their indifference curve where the wage share equals $m^{**} = \varrho/\nu$.[7] Regardless of the manner in which wages are determined, the freedom of firms to determine the rate of investment suffices to maintain shareholders' current consumption share constant. In the absence of government intervention, wage earners can only increase wages at the expense of investment and, hence, of future wage increases. Wage earners are structurally dependent.

Structural Dependence of the State

Now let us consider a government that taxes and transfers income. The government imposes a tax either on income from capital or on wages and transfers the collected revenue to the other group. Taxes, T, are collected according to $T = t(1 - m)Y$, where t is the tax rate. If the tax rate t is positive, the tax is imposed on income from capital, and the revenue is transferred to wage earners. If the tax rate t is negative, the tax is imposed on wages, and the revenue is transferred to firms.[8]

Wage earners' share of national income is now $(W + T)/Y$, while the amount of net profits is

$$P = (1 - m)Y - I - T$$
$$= (1 - m)(1 - s - t)Y.$$

The rate of investment firms choose now, given the tax rate and the share of wages in income, is

$$s^*(m,t) = (1 - t) - \frac{\varrho/\nu}{1 - m}$$
$$= s^M - t.$$

With a tax on income from capital, tax payments come entirely out of investments. With a tax on wages, tax payments go entirely into investments. Shareholders continue to consume the same share of income as before, since $[1 - t - s^*(m,t)] = (1 - s^M)$. The share of income that is invested with taxes is $I/Y = 1 - m - \varrho/\nu - (1 - m)t$.

Thus the trade-off faced by the government between redistribution of income through taxation and investment is identical to the trade-off between wages and investment: $d(I/Y)/d(T/Y) = -1$. We can relabel the horizontal axis of Figure 1 to read $(W + T)/Y = m + t(1 - m)$, and use the same line to describe the possibilities facing the government as it raises or lowers the tax rate. The state is in the same situation as wage earners are.

What will different governments do when confronted with this trade-off? Because shareholders' welfare declines as the tax rate increases, $dP^*/dt < 0$, a pro-business government would tax wage earners and transfer the income to firms: it would generate growth financed by workers. A purely procapitalist government, a government which sought to maximize P^*, would tax wages at the maximum feasible rate. A purely prolabor government, on the other hand, which sought to maximize W^* would have the same indifference curves as wage earners in Figure 1. Such a government would choose the tax rate such that $(W + T)/Y = m + t(1 - m) = \varrho/\nu$, or $t = (\varrho/\nu - m)/(1 - m)$.[9]

Thus procapital governments favor growth over wage earners' consumption while prolabor governments prefer greater consumption for wage earners at the cost of slower growth. No government, however, can reduce the share of income that owners of capital consume. Any additional income for wage earners, whether it consists of wage gains won at the bargaining table or of transfer payments won through elections, reduces

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total investment, dollar for dollar. As long as $\varrho/\nu < 1$, wage earners will moderate their claims on profits and so will governments representing wage earners.

Given this conclusion, it is not surprising that democracy appears so feeble when faced with the power of capital. "The institutional *form* of this state," write Offe and Runge (in Offe 1984, 121), "is determined through the rules of democratic and representative government, while the material *content* of state power is conditioned by the continuous requirements of the accumulation process." "The presumed sovereignty of the democratic citizenry," Bowles and Gintis (1986, 90) state succinctly, "fails in the presence of capital strike." The democratic process may bring governments with radically different bases of support into office but not governments with radically different programs.

Is the State Structurally Dependent on Capital? Statics

Taxes on Consumption out of Income from Capital

These conclusions are based, however, on an assumption that need not be true of capitalism in general. This assumption is that the only policy instrument governments can possibly use is a tax on income. This particular policy instrument leaves the state structurally dependent because utility-maximizing shareholders respond to tax increases with less investment. But governments can adopt policies that alter the trade-off between investment and income distribution. The inference that because society is dependent on capital, so must be the state is false.

In fact, all governments in developed capitalist countries tax, at different rates, the part of income from capital that is invested and the part that is consumed. Tax incentives for investment have been widely used, at least since the end of World War II. Depreciation allowances that differ from the rate at which plants and equipment actually wear out or become obsolete, investment credits, investment grants, special treatment of capital gains, and double taxation of profits distributed as dividends are only some of the most common deviations from a straight income tax. (For recent reviews of investment incentives in different countries, see Bracewell-Milnes 1980; Bracewell-Milnes and Huiskamp 1977; Commission of the European Community 1975; Hogan 1967; King and Fullerton 1984; Kopits 1981; Organization for Economic Cooperation and Development [OECD] 1979, 1983, 1984; Price, Waterhouse & Co. 1981.)

Let us distinguish the two taxes. The tax imposed on all income from capital (gross profits) is $T_i = t_i(I + P) = t_i(1 - m)Y$. We will refer to it as the "income tax." This is the tax we analyzed in the preceding section.

The tax imposed on shareholders' consumption is $T_c = t_cP = t_c(1 - s)(1 - m)Y$. This is the "consumption tax."[10]

Suppose now that a government adopts a pure consumption tax, that is, $t_i = 0$ so that $t_c(1 - s)$ is the tax rate applied to the income from capital $(1 - m)Y$. The most important effect of using this instrument is that the rate of investment is not affected by this tax rate, whatever it is, as long as wages are constant (Auerbach 1983). Indeed, as shown in the Appendix, for any rate of taxation of consumption out of profits, firms continue to invest at the market rate given by Equation 2:

$$s^*(m) = 1 - \frac{\varrho/\nu}{1 - m} = s^M.$$ [11]

We are thus already one step away from structural dependence: with this tax instrument a government can keep investment at its market determined rate and

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distribute the remaining income of shareholders to wage earners. A pure prolabor government can set t_c close to 1, reduce shareholders' share of consumption almost to 0, yet keep the rate of investment unchanged at s^M.

This much is true when the wage share is fixed. But in some capitalist countries wage earners are highly organized in centralized union federations with the capacity to shape the private distribution of income. If a government uses a pure consumption tax, the union federation will maximize W^* by choosing the share of income given by $m^{**}(t_c) = (1 - t_c)\varrho/\nu$. The willingness of the unions to engage in the political exchange described by Pizzorno (1978), Hibbs (1978), and Korpi and Shalev (1980), in which lower wages are traded for increases in welfare programs is apparent: as the tax rate, and thus transfers of income, increase, the federation reduces its preferred private wage share.

Because consumption taxes do not affect the rate of investment and do reduce the share of wages that centralized unions will demand and because the rate of investment increases as the share of wages declines (Equation 2), increases in the tax on consumption out of profit can increase the rate of investment. The rate of investment resulting from a tax on consumption of income from capital when wages are set unilaterally by centralized unions will be

$$s^*[m^{**}(t_c)] = 1 - \frac{\varrho/\nu}{1 - (1 - t_c)\varrho/\nu},$$

which is a positive function of t_c, $ds^*/dt_c > 0$.[12] This, then, is the second step: in concertation with the union federation the government can increase investment.

Can it do both: reduce the income share consumed by shareholders and increase investment above market-determined levels? The answer is positive; but because an even stronger conclusion can be sustained, let us go directly there.

Workers' Welfare under Socialism and Democratic Capitalism

The strong result is a vindication of the traditional social democratic strategy, formulated perhaps for the first time by Bernstein in 1898, elaborated by Henrik de Man in 1933, adopted by the Swedish Social Democrats in the 1930s, and codified in the German Social Democratic Programme of Bad Godesberg in 1959, namely, that a proworker state can control the capitalist economy in a manner that would bring the material welfare of wage earners to the same level they could reach under socialism.

Under capitalism, owners of capital stock control investment, while wage earners may at most control the distribution of private income. The strategic problem organized workers confront under capitalism is thus to choose the share of income, given that someone else determines investment. Under socialism, wage earners would control both the share of income going to investment and that going to consumption. Since capital stock would be collectively owned, there would no profit incomes; and the society as a whole, through some preference-revealing mechanism, would decide the distribution of income into investment and consumption.[13]

Wage earners' strategic problem under socialism would thus be to choose their share of income, knowing that whatever income they do not consume is invested. Technically, this is the same as setting $s = 1$ and choosing m to maximize W^*, which yields $m^*(1) = \varrho/\nu$. The rest of income goes to investment, or $I/Y = 1 - \varrho/\nu$. Associated with this allocation of resources is the level of wage earners' welfare under socialism $W^*(S)$. Given the technology and the time preferences, $W^*(S)$ represents the highest possible level

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of wage earners' welfare—the level they could attain if they controlled investment as well as income distribution.

Back to capitalism: consider the case with one big union and a government that chooses a pure consumption tax with the tax rate t_c close to 1. As t_c approaches 1,

$$(W + T)/Y = [m^{**} + t_c(1 - s^*)(1 - m^{**})] \rightarrow \varrho/\nu$$

and

$$I/Y = s^*(1 - m^{**}) \rightarrow 1 - \varrho/\nu,$$

which is the same as under socialism. As t_c approaches 1, wage earners' welfare under capitalism approaches the welfare they could obtain under socialism (Wallerstein and Przeworski, n.d.).

Would a government ever choose such a tax rate? To answer the question we need a language to characterize governments. Suppose the government's objective is to maximize G^*, which depends on the welfare of wage earners and shareholders according to $G^* = (W^*)^\gamma (P^*)^{1-\gamma}$, $0 < \gamma < 1$. Thus γ is the weight attached to the welfare of wage earners and $(1 - \gamma)$ is the weight given the welfare of shareholders. If γ approaches 1, labor's interests predominate; if γ approaches 0, the government exclusively favors capital.

The government's optimal tax on consumption out of income from capital is given implicitly by the equation

$$t_c = \frac{\gamma P^*}{\gamma P^* + (1 - \gamma)W^*}.^{14} \qquad (3)$$

Thus, as long as $0 < \gamma < 1$, the government will choose to tax consumption out of income from capital at a rate also between 0 and 1. In addition, the greater the weight given to wage earners' welfare, the higher will be t_c, or $dt_c/d\gamma > 0$. Finally, as γ approaches 1, t_c also approaches 1 and wage earners' welfare approaches that associated with socialism. Note that the share of national income that shareholders consume approaches 0, but shareholders' consumption in absolute terms may remain substantial. The conclusion is that when all wage earners are organized in one centralized union federation and the government is purely prolabor it will choose a tax on capitalist consumption the effect of which will be to bring wage earners' material welfare almost to the level they could obtain under socialism. Hence, at least when the union federation is willing to engage in a political exchange of private and public income, the state is not structurally dependent on capital.[15]

Caveats and Preliminary Conclusions

These conclusions are subject to a number of qualifications: (1) the results may depend on restrictive assumptions about workers' and shareholders' preferences; (2) structural dependence may work through mechanisms other than investment; (3) the state may be dependent not on national but foreign investors; (4) interventionist governments may be inherently incapable of serving as perfect agents of any group's interests. Let us consider each in turn.

For the sake of mathematical tractability, we have worked with particularly simple specifications of the production and utility functions. Our conclusion does not depend on the assumption of a linear production function. Auerbach (1983) and Przeworski and Wallerstein (1985) present models of investment-neutral taxes on profits using the standard production function with diminishing returns to investment. Nor is our conclusion restricted to the case of logarithmic utility functions. The same results can be derived with any utility function with a constant elasticity of marginal utility.[16] We do not know, however, the generality of our results within the entire set of quasi-concave utility functions with which economists like to work.

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Our results are sensitive to the assumption that firms care only about the present and future consumption of their shareholders. Owners of firms may not like encroachments on their prerogatives or the paying of taxes per se. Yet, the fact that owners of firms do not like something the government is doing, even if it touches them as profoundly as, say, a law increasing workers' rights at the workplace, is not sufficient to affect investment. Suppose firms maximize some more general utility function $V(P^*,x)$ where P^* is, as above, the welfare derived from present and future consumption and x is anything else that causes distress ($\partial V/\partial x < 0$) as the Left comes to power. "Investment strikes" will occur only if $dx/ds > 0$. Note that control over the workplace will not do for x, because it is independent of investment. Dislike of paying taxes will cause firms to increase, not decrease, their rate of investment in the presence of taxes on shareholders' consumption. Thus, likely candidates for x are not obvious.

Our results are also sensitive to the assumption that workers care only about their consumption. If workers determine the supply of labor by balancing their marginal utility of leisure with their marginal utility of income, the replacement of private wages with transfer payments financed by a tax on consumption out of profits will reduce incentives to work as it increases incentives to invest.

The second objection is that our conclusion is valid at most with regard to a specific mechanism of structural dependence, namely, the dependence of future material welfare on investment. One might postulate other mechanisms that would render the state dependent on capital, for example, the need to finance state activities through borrowing (see Marx 1952b, written in 1850). The mechanism most often suggested by Marxist theoreticians is the following: governments depend on current levels of economic activity, therefore they depend on investment. This formulation is, however, incoherent: if governments are concerned with current levels of activity, current demand, or current employment, they can and would do many things other than worry about investment. In a Keynesian textbook they can reduce taxes on wages, increase public employment, or print money as well as reduce interest rates to stimulate investment. A concern with the short run would not lead to a preoccupation with investment. Investment matters mostly for the long run.

In general, however, it may be true that the state is structurally dependent through mechanisms other than the one analyzed here. A reasonable short-term model of dependence is yet to be developed, but our conclusion should not be taken to be more general than it is.

Third is the issue of international constraints. A clarification may be needed. In the analysis, export of capital was treated as equivalent to consumption from the point of view of the tax system: whatever is not invested in the particular economy is subject to the tax. Thus all that governments need to do, in order to use the tax system to induce domestic (as distinct from foreign) investment, is to ascertain the actual amount of domestic income and investment. And in spite of transfer pricing and other mechanisms by which firms can shift tax burdens from country to country, there seems to be a general agreement that by and large, governments do monitor international flows of income (Bracewell-Milnes 1980; Hartman 1985; OECD 1983). Governments that seek to attract new foreign investment are in a different and more constrained situation. High taxes on consumption out of profits that have no effect on the investment decisions of domestic firms (or of local subsidiaries of multinational firms) will deter new foreign investors. The most important and unresolved issue, however, concerns the range of choice that the government of any country has when domestic

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producers must trade in international markets in which competitiveness depends to a significant degree on relative labor costs.

Finally, we come to the problem of the limits of governmental competence. Government intervention in the economy creates a potential for vast inefficiencies in the allocation of investment and labor. Any system of investment incentives is vulnerable to inefficiencies due to differing effective rates on different assets (King and Fullerton 1984). Any system in which private consumption depends primarily on transfers rather than wages is vulnerable to inefficiencies due to misallocation of labor. The problem lies not only in the difficulty of designing neutral taxes or transfers but in their implementation. The very fact that the state acquires the capacity to intervene in the economy, that is, to affect material welfare of private actors, turns it into an attractive target of influence by these actors. What has thus developed as the eventual aftermath of the Keynesian revolution is a state that is strong in its capacity to intervene and that is highly vulnerable to private influence (Schmitter 1985, Skidelsky 1977). Diagnoses formulated within ideologically distant perspectives coincide in their antistatist conclusions: Habermas's (1975) analysis of the crisis facing capitalism is identical to that of Stigler (1975). What may be thus wrong with our analysis is the assumption that interventionist governments can ever serve as perfect agents of the political forces that placed them in office. The economy is sick, the drugs are available, but the doctor may be a hack.

In sum, there may be limits to the ability of the state to redistribute consumption without discouraging private investment, because the economy depends on an inflow of capital from abroad (rather than the reinvestment of profit earned at home) or because private wages cannot be replaced by public income without undermining work incentives or because the government cannot be trusted with the discretion required by redistributive policies. But all of these reasons are less general than the claims of the theory of structural dependence.

Our analysis reveals a large range of economically feasible policies. Exclusive reliance on income taxes, that is, taxes that are independent of the rate of investment—whether these are imposed on capital or labor—allows owners of capital to choose the share of national income they want to consume and to ward off, by reducing the share of investment, all attempts both by unions and by governments to redistribute consumption. Under exclusive reliance on income taxes, the consumption of all groups other than shareholders, government included, is thus subjected to "market discipline." In contrast, if only consumption out of income from capital is taxed, governments can redistribute consumption without reducing investment. Acting in concert with unions willing and able to trade off wage demands for increased transfer payments, the government can increase investment and reduce capitalists' consumption share simultaneously.

Governments can choose different policies concerning distribution and growth and these policies can be effective. Specifically, governments can increase investment by reducing wage earners' consumption or by reducing the consumption of owners of capital. Since all social groups will seek to shift the costs of investment onto others, the outcome of the democratic process can matter greatly for the welfare of different groups under capitalism. The state is not structurally dependent.

These conclusions may seem banal, but they run against a substantial body of established theory. The Marxist theory presented in the first section is logically persuasive and historically plausible. Moreover, we demonstrated in the second section that as long as governments limit

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themselves, or are limited, to using a flat income tax, the state is structurally dependent on capital. And, importantly, Marxists are not the only theoreticians who come to the conclusion that the state is dependent: the language is different but this is the general thrust of theories of the state originating from the neoclassical economic theory, as exemplified by Peltzman (1976), Aumann and Kurz (1977), Busch and Mackay (1977), Becker (1983), Bates and Lien (1985), and innumerable others who conclude that all governments seeking support must anticipate the negative effect of taxes on private investment. Thus, there are good reasons to be theoretically surprised.

Is the State Structurally Dependent on Capital? Dynamics

Once they are in office, prolabor governments can use tax policies to bring the welfare of wage earners arbitrarily close to the level wage earners could attain if they controlled investment directly: this is the central conclusion we have reached thus far. This conclusion, however, holds only once the prolabor government is in office—more precisely, once the policy is in effect. The question that remains is what happens to investment and wage earners' welfare when firms anticipate that a prolabor party would or might win the next election and institute such tax policies.

Since the mathematical analysis is quite complicated, we present only the framework of analysis and the conclusions. Suppose that at time $\tau = 0$ a government characterized by γ_0 is in office and the tax rate is $t_c = t_0$. The government is highly procapitalist and this tax rate is close to 0. An election is to take place and firms expect that this election will be won by a prolabor government characterized by $\gamma_1 > \gamma_0$, which will eventually introduce, at time $\tau = 1$, a tax $t_c = t_1 > t_0$.[17] What rate of investment will be chosen during the period from $0 \leqslant \tau < 1$ if firms expect that tax rate to be higher at and after $\tau = 1$.

When firms do not anticipate a change in tax policy, they choose the rate of investment that maximizes $P^*(t_0)$ between $\tau = 0$ and infinity. If, however, they anticipate that the tax will increase at $\tau = 1$ and remain at the new level, they will choose a time path of investment $s(\tau)$ to maximize

$$\int_0^1 e^{-\varrho\tau}\ln[P(t_0, \tau)]d\tau + \int_1^\infty e^{-\varrho\tau}\ln[P(t_1, \tau)]d\tau.$$

The conclusion is the following: firms that anticipate a future increase in taxes will reduce investment in the period $0 \leqslant \tau < 1$ even if the tax is a pure consumption tax that will not lower investment in the period $\tau \geqslant 1$. The same holds for firms that are not certain about future tax policy but think taxes will rise with some positive probability. Thus, anticipations of certain or possible increases of taxes lead firms to lower their investment rate. Similarly, expectations of future tax decreases will cause current investment to increase. Pure consumption taxes are no longer investment-neutral when change is anticipated.

Thus "business confidence"—the expectation by firms concerning government policy—has direct consequences for the investment behavior of firms. Yet this conclusion should be distinguished from the view of Kalecki (1966; also Block 1977), who also saw in business confidence the constraint that any government under capitalism is forced to consider. In Kalecki's view, business confidence threatened investment even in a static case, when the government policy was already in place and no changes were anticipated. However, once a government

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has implemented a consumption tax, the investment rate is unaffected by the tax rate. It is the anticipation that a government would introduce or raise such a tax that erodes business confidence and lowers investment.[18]

We suspect that many newly elected proworker governments abandoned or moderated their programs for wrong reasons, having mistaken the cost of anticipation of policies for the costs of policies themselves. The French Socialist party announced in the electoral campaign of 1981 that it would propose a law in favor of renters. Once elected, it prepared such a bill. In the meantime, the vacancy rate went up and so did rents. Having listened to a chorus of right-wing press, the government decided that the policy was too expensive and moderated the provisions of the law. Our analysis suggests that this may have been a mistake. The government never waited to discover the costs of the law. Instead, it panicked over the costs of anticipation which, by then, were sunk.

Ever since Winston Churchill sneered in 1924 that the Labour party was not fit to govern, parties of labor have been making desperate efforts to show their moderation and responsibility in administering the economy once in office. Yet once the decision to adopt a policy has been made and once the government has suffered the effects of anticipation, the government is best-off introducing the original program without any compromises. The time to compromise is not after the costs have already been paid.[19]

Since anticipated transitions to higher taxes are costly for wage earners' welfare, socialists are sometimes tempted to proceed "in one stroke"—to use the phrase of Oskar Lange, who thought that if a government has decided to go ahead with a socialist program at all, "it must go through resolutely . . . at a maximum speed" (1964, 125; also Kautsky 1925). This is not a new idea: Machiavelli advised, "When a prince takes a new state, he should calculate the sum of all the injuries he will have to do, and do them all at once, so as not to have to do new ones every day; simply by not repeating them, he will thus be able to reassure people, and win them over to his side with benefits" (1977, 28).

Our analysis suggests that socialist governments should never announce more than they intend to do and that they should do what they announced fully and quickly. Leon Blum was wrong to have waited to assume office after the Popular Front won the elections of 1936. Yet we do not know whether one stroke is better than many. Indeed, we cannot even determine whether or not workers' parties should seek to increase the tax on shareholders' consumption. We know that once instituted, an increase in t_c will increase wage earners' welfare until or unless the socialist government is defeated and the policy reversed. But we also know that anticipation of this moment causes a fall in investment and is costly to wage earners' welfare in the period before the tax increase goes into effect. Thus, the question remains open whether the welfare of wage earners is improved. If workers are better-off when prolabor governments adhere to the status quo, then the state is structurally dependent in dynamic terms: some distributions of consumption are unattainable because the transition costs are too high.

Structural Dependence and Socialist Policies

The capitalist economy, in which owners of wealth and of the capacity to work make decentralized decisions about allocations of their endowments, regularly generates a number of effects that are experienced as profound deprivation by large segments of society. The market does not assure the material security of

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anyone who does not own wealth and is not able to earn a living. The market generates drastic inequalities, including inequality of opportunity. Finally, in cases where the private rate of return diverges from the social one, the market allocates resources to uses inefficiently. This has been the traditional socialist view of capitalism.

The perennial question has been what, if anything, can be done about it. Social Democrats in several countries developed a full-fledged program for steering the capitalist economy in the interest of people who depend on income from employment or cannot earn an income sufficient to survive. By controlling the capitalist economy, the state run by socialists could rationalize the economy as a whole and orient it toward the general welfare, the Swedish Social Democratic minister of finance discovered in the 1930s (Wigforss, as summarized by Lewin 1975, 286). Because capitalism became organized, it could be administered by socialists in the interest of the people, argued the eventual German Social Democratic minister of finance (Hilferding 1981). Ironically, it was Keynes who provided the policy instruments for controlling capitalist economies and perhaps the most eloquent statement of the social-democratic position: "It is not the ownership of the instruments of production which it is important for the state to assume. If the state is able to determine the aggregate amount of resources devoted to augmenting the instruments and the basic reward to those who own them, it will have accomplished all that is necessary" (1964, 378).

Even in its heyday, however, the social-democratic strategy was the subject of at least skepticism if not hostility from those who were persuaded that no government, not even one elected by "the people," could control investment and income distribution as long as the instruments of production were privately owned. The thesis that the state is structurally dependent on capital was used by critics of social democracy to argue that this strategy is unfeasible, since the ability of any government to determine the level of investment and the distribution of income is subject to a trade-off given by the self-interest of owners of the instruments of production.

Our contribution to this debate is as follows: Once in place, a tax on shareholders' consumption does not reduce investment. The state is not structurally dependent on capital in the sense that virtually any distribution of consumption between wage earners and shareholders is compatible with continual private investment. Yet anticipations of future threats to shareholders' consumption do reduce investment and impose costs on wage earners. Whether and when the future benefits to wage earners outweigh the costs incurred during the transition period remains an open question.

Appendix

Preliminaries

Given that the economy grows according to $Y'(\tau) = \nu s(1 - m)Y(\tau)$, the national income at any time is given by $Y(\tau) = Y(0)e^{\nu s(1-m)\tau}$. Wages, profits, investment, and taxes all constitute shares of this income as described in the text. All of our results are derived from the first-order conditions of a series of maximization problems. In all cases, it is easily verified that the second-order conditions for a maximum are satisfied.

Capitalists' Problem

Firms choose s to maximize

$$P^* = \int_0^\infty e^{-\varrho\tau}\ln[P(\tau)]d\tau = \frac{1}{\varrho}\left[\ln P(0) + \frac{\nu s(1-m)}{\varrho}\right]. \quad \text{(A-1)}$$

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Note that P^* is only well defined when $P(0) > 0$. To insure that this condition is met, we assume throughout that $m < 1$ or $\varrho/\nu < 1$.

Without taxes, $P(0) = (1 - s)(1 - m)Y(0)$. Firms thus choose the s that satisfies

$$\frac{\partial P^*(s,m)}{\partial s} = \frac{1}{\varrho}\left[\frac{\nu(1-m)}{\varrho} - \frac{1}{1-s}\right] = 0$$

which implies the choice

$$s^*(m) = 1 - \frac{\varrho/\nu}{(1-m)} = s^M. \qquad \text{(A-2)}$$

Capitalists are consuming the share $(1 - s^M)(1 - m) = \varrho/\nu$ and investing the share $s^M(1 - m) = (1 - m - \varrho/\nu)$.

With income and consumption taxes,

$$P(0) = [(1 - t_c)(1 - s) - t_i](1 - m)Y(0).$$

The first-order condition is now

$$\frac{\partial P^*}{\partial s} = \frac{1}{\varrho}\left[\frac{\nu(1-m)}{\varrho} - \frac{1-t_c}{(1-t_c)(1-s) - t_i}\right] = 0,$$

which yields

$$s^*(m,t_i,t_c) = 1 - \frac{\varrho/\nu}{(1-m)} - \frac{t_i}{(1-t_c)}. \qquad \text{(A-3)}$$

In the case with only income taxes, $t_c = 0$ and Equation A-3 reduces to

$$s^*(m,t_i) = 1 - \frac{\varrho/\nu}{(1-m)} - t_i = s^M - t_i. \qquad \text{(A-4)}$$

Capitalists are consuming the share $(1 - s^* - t_i)(1 - m) = \varrho/\nu$ and investing the share $s^*(1 - m) = [(1 - m)(1 - t_i) - \varrho/\nu]$.

In the case with only consumption taxes, $t_i = 0$ and Equation A-3 reduces to

$$s^*(m,t_c) = 1 - \frac{\varrho/\nu}{(1-m)} = s^M. \qquad \text{(A-5)}$$

Now capitalists are consuming the share $(1 - t_c)(1 - s^*)(1 - m) = (1 - t_c)\varrho/\nu$ and investing the share $s^*(1 - m) = (1 - m - \varrho/\nu)$. As t_c goes to 1, capitalists' consumption share goes to 0.

Workers' Problem

Unions choose m to maximize

$$W^* = \int_0^\infty e^{-\varrho\tau}\ln[W(\tau) + T(\tau)]d\tau = \frac{1}{\varrho}\left[\ln[W(0) + T(0)] + \frac{\nu s(1-m)}{\varrho}\right]. \qquad \text{(A-6)}$$

If there are no taxes and the unions treat s as exogenous, $W(0) = mY(0)$ and the solution to Equation A-6 (for $s > 0$) is given by

$$\frac{\partial W^*(m,s)}{\partial m} = (1/\varrho)[(1/m) - (\nu s/\varrho)] = 0,$$

which implies $m^*(s) = \varrho/\nu s$.

If firms invest according to their best reply $s^*(m)$ given by Equation A-2, $\nu s^*(1 - m)/\varrho = [\nu(1 - m)/\varrho - 1]$ in Equation A-6 and the solution to the workers' problem becomes

$$\frac{dW^*[m,s^*(m)]}{dm} = (1/\varrho)[(1/m) - (\nu/\varrho)] = 0$$

with $m^{**} = \varrho/\nu$.

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With income and consumption taxes, $[W(0) + T(0)] = \{m + [t_i + t_c(1 - s)](1 - m)\}Y(0)$. If unions anticipate capitalists' response $s^*(m, t_i, t_c)$ as given by Equation A-3, the workers' solution is given by

$$\frac{dW^*}{dm} = \frac{1}{\varrho}\left[\frac{1 - A}{m + (1 - m)A + t_c\varrho/\nu} - \frac{1 - A}{\varrho/\nu}\right] = 0 \qquad \text{(A-7)}$$

where $A = t_i/(1 - t_c)$. Solving for m^{**},

$$m^{**}(t_i, t_c) = \frac{(1 - t_c)^2\varrho/\nu - t_i}{1 - t_c - t_i}. \qquad \text{(A-8)}$$

Note that when $s = s^*(m, t_c, t_i)$, workers' share $(W + T)/Y = [m + (1 - m)t_i/(1 - t_c) + t_c\varrho/\nu]$. Therefore it follows immediately from Equation A-7 that when workers anticipate capitalists' best response s^* in choosing m, workers consume the share ϱ/ν of national income.

In the case with only income taxes, $t_c = 0$ and Equation A-8 reduces to $m^{**}(t_i) = (\varrho/\nu - t_i)/(1 - t_i)$. In the case with only consumption taxes, $t_i = 0$ and Equation A-8 reduces to

$$m^{**}(t_c) = (1 - t_c)\varrho/\nu. \qquad \text{(A-9)}$$

The Government's Problem

The government's objective function depends on the welfare of workers and capitalists according to the equation $G^* = (W^*)^\gamma(P^*)^{1-\gamma}$, $0 < \gamma < 1$. To find the government's optimal tax, we take the derivative

$$\frac{dG^*}{dt_j} = (W^*)^{\gamma-1}(P^*)^{-\gamma}\left[\gamma(P^*)\frac{dW^*}{dt_j} + (1 - \gamma)(W^*)\frac{dP^*}{dt_j}\right] = 0 \qquad \text{(A-10)}$$

where t_j represents either t_i or t_c. Two cases were discussed in the text.

First we consider the income tax that would be chosen by a purely proworker government. Note that if $\gamma = 1$, then $G^* = W^*$. Thus, we substitute $s^*(m, t_i)$ (Equation A-4) into W^* (Equation A-6) and differentiate to obtain

$$\frac{dW^*}{dt_i} = \frac{(1 - m)}{\varrho}\left[\frac{1}{t_i(1 - m) + m} - \frac{\nu}{\varrho}\right] = 0,$$

which can be simplified to yield the solution $t_i = (\varrho/\nu - m)/(1 - m)$.

Now consider the case with a pure consumption tax, $t_i = 0$, and with unions choosing $m = m^{**}$. Substitute $m^{**}(t_c)$ (Equation A-9) and $s^*(m^{**}, t_c)$ (Equation A-5) into W^* (Equation A-6) and take the derivative

$$dW^*/dt_c = 1/\varrho \qquad \text{(A-11)}$$

Similarly, substitute $m^{**}(t_c)$ and $s^*(m^{**}, t_c)$ into P^* (Equation A-1) and differentiate

$$dP^*/dt_c = -t_c/\varrho(1 - t_c). \qquad \text{(A-12)}$$

Combining Equations A-11 and A-12 with A-10 and rearranging terms yields $\gamma(1 - t_c)P^* - (1 - \gamma)t_cW^* = 0$, or

$$t_c = \frac{\gamma P^*}{\gamma P^* + (1 - \gamma)W^*}, \qquad \text{(A-13)}$$

which implicitly defines the optimal t_c (since both P^* and W^* contain t_c). Thus, $0 < t_c < 1$ (since $0 < \gamma < 1$). It is readily checked that W^* does not explode as γ approaches 1. Therefore, as γ approaches 1, $(1 - \gamma)W^*$ approaches 0 and t_c approaches 1. Similarly, it is easily checked that P^* does not explode as γ approaches 0. Thus as γ goes to 0, γP^* goes to 0 and so does t_c. To find $dt_c/d\gamma$, differentiate Equation A-13 implicitly and rearrange terms to write

$$\frac{dt_c}{d\gamma} = \frac{(1 - t_c)p^* + t_cW^*}{\gamma P^* + (1 - \gamma)W^* + t_c/\varrho} > 0.$$

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Notes

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1. These figures are rough estimates for the postwar U.S. economy. Estimates of the productivity of capital can be found in Hill 1979. The others can be found in the National Income and Product Accounts published by the Department of Commerce. Of course, investment in physical capital is only one source of economic growth. We emphasize the role of investment in growth because that is the way the theory of structural dependence has been formulated.

2. Alternatively and equivalently, firms maximize gross profits and distribute all income accruing to capital to shareholders, who then choose the rate of savings that maximizes their utility (see Hirshleifer 1970).

3. We use the log function to simplify the presentation, but all of our results hold for the entire class of utility functions with a constant elasticity of marginal utility.

4. Here we consider the rate of time preference ϱ to be risk-free and thus the same for all individuals who derive utility from consumption, whether they are workers or capitalists. We took a different approach in our earlier analysis of class conflict (Przeworski and Wallerstein 1982a), where rates of time preference reflected the differential risk faced by owners of firms and wage earners. Given the problem at hand, it seems better to consider time preferences risk-free and to model uncertainty explicitly. The effects of uncertainty are discussed in the final section.

5. When workers' share m is greater than or equal to unity, profits are negative and firms will disinvest at the maximal rate.

6. We treat investment in neoclassical fashion as the result of an intertemporal allocation of consumption by owners of capital. The Keynesian approach, in contrast, emphasizes the importance of investing to satisfy future demand. In a Keynesian model, the derivative ds^*/dm would be positive since wage increases put more money in the hands of people with a high propensity to consume. The problem of structural dependence cannot arise within a Keynesian framework where increasing wage earners' consumption accelerates investment and may even increase current profits. Indeed, one of the political appeals of Keynesian theory was the denial of any conflict of interests between wage earners and capitalists (see Skidelsky 1977 and Przeworski and Wallerstein 1982b).

7. In technical language, workers are behaving as Stackelberg leaders, firms as Stackelberg followers. This means that the unions anticipate the effect of their choice of wage share on firms' investment decisions, while firms respond optimally to whatever unions choose. We do not, in this paper, consider the alternative of firms leading and unions following, although we did previously in Przeworski and Wallerstein 1982a. The role of Stackelberg leaders poses a collective-action problem for firms. To act as Stackelberg leaders would demand that firms collectively invest more than is optimal for each individually.

We use the double asterisk to denote the optimal choice of the Stackelberg leader and the single asterisk to denote the best reply strategy or the optimal choice of the Stackelberg follower.

8. In this case, the rate at which wages are taxed is given by the equality $T = -t(1 - m)Y = t_w mY$, which implies that the tax rate imposed on wages is $t_w = -t(1 - m)/m$.

9. Wage earners who acquire complete control over tax policy through electoral victories achieve no more and no less than wage earners who gain complete control over private income distribution through collective bargaining: the outcome is the same in both cases. Note that the wage earners' optimal tax may be negative.

10. The total tax rate imposed on income from capital is thus $t = t_i + t_c(1 - s) = (t_i + t_c) - t_c s = t_0 - t_1 s$, where $t_0 = t_i + t_c$ is the nominal tax rate and $t_1 = t_c$ is the rate of investment relief.

11. This result may be nonintuitive: firms pay higher taxes and invest at the same rate as before. The reason is that this tax imposes the same cost on consumption now and in the future and thus does not change the optimal allocation of consumption over time.

12. This follows as long as the government chooses the tax rate t_c such that $(1 - t_c)\varrho/\nu < 1$. Otherwise firms will disinvest at the maximal rate no matter what the government does.

13. Obviously this notion of socialism should not be confused with Eastern European economies, where the decisions about allocation of resources are made by party bureaucrats.

14. Both P^* and W^* contain t_c so, contrary to appearances, t_c is not isolated on the left-hand side of Equation 3.

15. Whether a prolabor government in countries where unions are fragmented or weak could still bring wage earners' welfare arbitrarily close to the level they could attain under socialism with an appropriate set of taxes and transfers remains to be investigated.

16. The family of utility functions with a constant elasticity of marginal utility is conventionally given by the formula $U(c) = [1/(1 - b)]c^{1-b} + B$, $b > 0$, for $b \neq 1$ and $U(c) = \ln(c)$ for $b = 1$. The parameter b is the elasticity of marginal utility.

17. It makes no difference in the analysis if firms are certain taxes will increase at $\tau = 1$ or attribute

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some positive probability to a tax increase at $\tau = 1$. Uncertainty can be understood in this context as a situation in which firms attach subjective probability distributions to the outcomes of future elections or directly to future tax rates. Exactly the same analysis would apply if firms are uncertain about the future wage share or the future productivity of capital.

18. Aubin and Goyeau (1986) have found that in France the prospect of an electoral victory of the Left (measured as the product of the support for the Left in the polls and the proportion of the electoral term that has elapsed) contributes significantly to the explanation of investment in a model that includes measures of aggregate demand and the cost of capital.

19. This does not mean, of course, that governments should stubbornly stick with all policies regardless of their effects. Some policies are simply bad policies.

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